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Fault Tolerant Control of the B747 Short-Period Mode using Progressive Accommodation

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Abstract— **Linear Quadratic design for the fault tolerant control of the short-period mode of the longitudinal model of the Boeing 747 aircraft is investigated. Under actuator faults, both fault accommodation and system reconfiguration procedures need the controller to be re-designed, which means solving the Algebraic Riccati Equation (ARE) associated with the post-fault system model. To overcome the risks of system instability and/or control inadmissibility, which result from fault detection and isolation delay, and computation time of the accommodated control, a progressive accommodation strategy, based on the Newton-Raphson algorithm for solving AREs, is used. The application shows the benefits associated with this approach.**

I. Introduction

Fault Tolerance (FT) has established itself as an important research area in the last decade [1], [2], [3]. It refers to the mechanism by which a given system is still able to exhibit desired properties, both in normal operation and in the presence of faults occurring in sensors, actuators, or in the plant [4]. Properties of interest follow from the design of the nominal control system and range from stability [5], disturbance attenuation [6], to model matching [7], and optimality [8]. FT mechanisms may be implemented either via pre-computed control laws, using multiple-model [9] or gain-scheduling [10] techniques, or via on-line automatic re-design, using linear quadratic regulation [11], model following [12], eigenstructure assignment [13], model predictive control [14], or pseudo-inverse [15] techniques.

Linear Quadratic (LQ) Control is one of the most widely applied modern control techniques. In the aeronautical field, developments on different aspects of the technique, as Loop Transfer Recovery (LTR) [16], or Linear Quadratic Output Feedback Optimal Control [17], have been reported in [18], [19], and [20].

In the state-space setting, the optimal control problem is defined by the transfer of the system state from one region of the state-space into another one in an optimal way, e.g. in minimum time, or with minimum energy. Besides, an important property of the Linear Quadratic Control technique is that it allows the designer to take into account both requirements on the amplitude of the control inputs and the settling time of the state variables.

In the Fault Tolerant Control (FTC) framework, the fault tolerance property of a control system means that the designed control law aims at keeping the system functionality available, by providing that a given objective can still be satisfied in the presence of faults which belong to a given set. However, most existing works deal with "ideal" systems, in which fault detection, isolation and estimation, and fault accommodation/system reconfiguration procedures, take no time, while in real life, the accommodated/reconfigured control law becomes available after some while, and there is a period during which the post-fault system is still controlled using the nominal control law. This may possibly lead to system instability and/or control inadmissibility of the accommodated/reconfigured control, which represent a serious problem for safety-critical systems, e.g. aircrafts, nuclear reactors, where the malfunctions may cause big human, economic, and environmental losses.

Precisely, in the LQFTC problem, after a fault, the new control law is designed by solving a new Algebraic Riccati Equation (ARE) [21], associated with the post-fault system model, which is a heavy and time-consuming task. The Newton-Raphson algorithm for solving AREs may represent an appropriate means of developing the iterative accommodation approach. In this paper, the progressive accommodation technique is applied for the fault tolerant control of the short-period longitudinal mode of a Boeing 747 aircraft. It is organized as follows : Section II introduces the nominal and the fault tolerant linear quadratic control problems. Section III presents the classical fault accommodation scheme versus the progressive one based on the Newton-Raphson algorithm. Section IV addresses the short-period mode control of the Boeing 747 aircraft longitudinal model and compares the results obtained, using the previous two different approaches. Section V concludes the paper.

II. Nominal and Fault Tolerant Linear Quadratic Control

Let subscripts n and f respectively stand for the nominal and the faulty system conditions.

A. Nominal Operation

Consider the linear time-invariant (LTI) deterministic system, whose nominal operation is modeled by :

$$M_n : \quad \dot{x}(t) = A_n x(t) + B_n u_n(t) \quad x(t_0) = x_0 \tag{1}$$

where $A_n \in R^{n\times n}$ and $B_n \in R^{n\times m}$ are the nominal system and control matrices, $x(t) \in R^n$ and $u(t) \in R^m$ are the state and control vectors, and $x_0 \in R^n$ is the initial condition.

Define the quadratic cost functional as :

$$J(x_0, u(t)) = \int_0^\infty \left[x^T(t)Qx(t) + u^T(t)Ru(t)\right] dt \tag{2}$$

where $R = R^T > 0$ and $Q = C^T C \geq 0$ are weighting symmetric matrices, and $(.)^T$ stands for transposition.

If the pairs (A_n, B_n) and (A_n, C) are stabilizable and detectable, the nominal linear state-feedback control law :

$$u_n(t) = -K_n x(t) \tag{3}$$

where $K_n \in R^{m\times n}$, will have the optimal form given by :

$$K_n = R^{-1} B_n^T P_n \tag{4}$$

Matrix P_n is the unique symmetric positive semi-definite solution of the Algebraic Riccati Equation (ARE) :

$$A_n^T P_n + P_n A_n + Q - P_n B_n R^{-1} B_n^T P_n = 0 \tag{5}$$

In the nominal case, the closed-loop system :

$$\dot{x}(t) = (A_n - B_n K_n) x(t) \tag{6}$$

is stable. Moreover, the control law (3) will provide the minimum value of criterion (2) :

$$J_{M_n}(x_0, u_n) = x_0^T P_n x_0 \tag{7}$$

B. Faulty Operation

When faults occur, the system to be controlled is described by the pair $(A_f, B_f) \neq (A_n, B_n)$, assuming that a LTI model is still able to capture the system behavior :

$$\begin{aligned} M_n : \quad \dot{x}(t) &= A_n x(t) + B_n u_n(t) \quad t \in [0, t_f[\\ M_f : \quad \dot{x}(t) &= A_f x(t) + B_f u_f(t) \quad t \in [t_f, \infty) \end{aligned} \tag{8}$$

There are two different means by which the fault can be recovered, according to the FT strategy that is used [22] :

1. In the *accommodation* strategy, the new control applies to the faulty system. Its model (A_f, B_f) must be identified on-line by the diagnostic algorithm, in addition to its fault detection and isolation functionalities.
2. In the *reconfiguration* strategy, faulty components are switched off and only the healthy part of the system is controlled. Its model (A_f, B_f) is therefore easily known, after fault detection and isolation has been performed.

When a fault occurs, stability is the primary requirement for the control system, and this differentiates four periods in the system operation :

- **fault-free** : $t \in \left[0, t_f\right[$: the system is in normal operation (model M_n), and the applied control is the nominal one, u_n.
- **fault detection** : $t \in \left[t_f, t_{fdi}\right[$: the system is faulty (model M_f), but the FDI algorithm has not yet detected, isolated and estimated the fault, and the control has not been accommodated, therefore the nominal control u_n is still applied.
- **computation of recovery control** : $t \in \left[t_{fdi}, t_{ftc}\right[$: the system is faulty (model M_f), the FDI algorithm has detected, isolated and estimated the fault, but the control has not yet been accommodated/reconfigured, therefore the nominal control u_n is still applied.
- **fault-recovered** : $t \in \left[t_{ftc}, \infty\right)$: the system is faulty (model M_f), the accommodated/reconfigured control u_f has been computed and applied.

1) Actuator Faults: For simplicity, only actuator faults are considered, i.e. the new LTI model of the system is described by the pair $(A_f = A_n, B_f \neq B_n)$.

Whatever the selected fault recovery technique, the best post-fault strategy consists of applying the optimal control solution to the system :

$$M_f : \quad \dot{x}(t) = A_n x(t) + B_f u_f(t) \tag{9}$$

with the initial condition $x_{t_f} = x(t_f)$, on the time interval $t \in [t_f, \infty)$. Finding the optimal control associated with the faulty system is a controller re-design problem, which consists of solving the ARE associated with the new situation, i.e. (5) where the nominal control matrix, B_n, is replaced by the post-fault one, B_f, namely :

$$A_n^T P_f + P_f A_n + Q - P_f B_f R^{-1} B_f^T P_f = 0 \tag{10}$$

Assuming that the pair (A_n, B_f) is still stabilizable, then the accommodated control becomes :

$$u_f(t) = -R^{-1} B_f^T P_f x(t) = -K_f x(t) \tag{11}$$

leading the criterion (2) to the value :

$$J_{M_n, M_f}(x_0, u_n, x_{t_f}, u_f) = x_0^T P_n x_0 + x_{t_f}^T (P_f - P_n) x_{t_f} \tag{12}$$

III. Classical vs. Progressive Fault Accommodation

A. Preliminaries

Equations (1) to (12) define LQFTC, i.e. the FTC problem in the LQ framework. However, the fact that the solution (u_n, u_f) achieves the objective in an optimal way does not mean that it is satisfactory. Indeed, $J_{M_n, M_f}(x_0, u_n, x_{t_f}, u_f)$ might be too high, or the behavior associated with the accommodated control might be unacceptable, thus denying system M_n to deserve the "fault tolerant" label with respect to the fault situation M_f. Let us introduce the following definitions.

Definition 1 (Admissibility): Assuming that a solution (u_n, u_f) to the FTC problem exists, it is admissible if and only if, for all initial conditions x_0 :

$$J_{M_n, M_f}(x_0, u_n, x_{t_f}, u_f) \leq \rho(x_0) J_{M_n}(x_0, u_n) \tag{13}$$

where $\rho(x_0) \geq 1$ is some predefined function which represents the maximal performance degradation that can be

admitted in faulty situations, for the initial condition x_0, and :

$$\Phi\left(M_n, M_f\right) \leq 0 \tag{14}$$

where Φ is a set of matrix inequalities that define admissible closed-loop behaviors.

Definition 2 (Fault Tolerance): System (1) is fault tolerant with respect to the fault situation described by (8), for x_0, if and only if the accommodation or the reconfiguration problem has an admissible solution.

The complete development of admissible solutions and system fault tolerance in the LQ framework against different choices of $\rho(x_0)$ can be found in [8], [23], while [24], [25] develop the presentation of FTC based on admissible models.

Classical development in solving ARE is based on the Schur method [26]. However, one of the best studied numerical methods remains the Newton method [27], which is presented next.

Theorem 1 (Kleinman, 1968): Let P_i, $i = 0, 1, \ldots$ be the unique positive semi-definite solution of the linear equation :

$$A_i^T P_i + P_i A_i + Q + K_i^T R K_i = 0 \tag{15}$$

where, recursively :

$$K_i = R^{-1} B^T P_{i-1}, \quad i = 1, 2, \ldots \tag{16}$$

$$A_i = A - BK_i \tag{17}$$

and K_0 is chosen such that the matrix $A - BK_0$ has eigenvalues with negative real parts. Then :

$$0 \leq P_N \leq P_{i+1} \leq P_i \leq \ldots \leq P_0, \quad i = 1, 2, \ldots \tag{18}$$

$$\lim_{i \to \infty} P_i = P_N \tag{19}$$

Different stabilizing procedures to provide the initial gain of the scheme have been reported in [28], [29], [30], [31].

From Theorem 1, it follows that, given any initial stabilizing matrix K_0, for N large enough, the solution of the linear algebraic Lyapunov equation :

$$A_N^T P_N + P_N A_N = -\left(Q + K_N^T R K_N\right) \tag{20}$$

converges to that of the algebraic Riccati equation :

$$A^T P_N + P_N A + Q - K_N^T R K_N = 0 \tag{21}$$

Replacing index N by f and considering $A = A_n$, one can see that (21) and (10) are identical.

B. Classical Accommodation Scheme

Assuming that N is the number of steps needed to obtain a satisfactory approximation for the solution of (20) and taking Δ the duration of one iteration, the classical control scheme would apply the nominal control law (3), during the period $t_{ftc} - t_{fdi} = N\Delta$, until the accommodated one (11) is available.

In this case, after the fault occurrence, the control sequence is defined by the following steps :

1) Phase 1. Computing the new control law :

$$\dot{x}(t) = (A_n - B_f K_n) x(t) \quad t \in [t_f, t_{fdi} + N\Delta[\tag{22}$$

2) Phase 2. The Newton-Raphson algorithm has converged :

$$\dot{x}(t) = (A_n - B_f K_N) x(t) \quad t \in [t_{fdi} + N\Delta, \infty) \tag{23}$$

C. Practical Aspects of Implementation

To analyze the efficiency of the classical accommodation scheme III-B, reconsider the value of criterion (12), in the faulty case, and remind the admissibility conditions (13) and (14), for the fault tolerant control solutions.

In the situation described in Section III-B, the computation delay is $t_{ftc} - t_{fdi} = N\Delta$. The total delay, $t_{ftc} - t_f$, is the sum of the fault detection, isolation, and estimation delay, $t_{fdi} - t_f$, and of the post-fault control accommodation delay, which is equal to $N\Delta$, $t_{ftc} - t_{fdi}$. Obviously, on the time interval $\left[t_f, t_{ftc}\right[$, the system control is inappropriate.

Precisely, let $\varphi(M_n, t)$ be the transition matrix associated with the nominal optimal system. Then, after the fault, the time evolution of the system state is governed by :

$$\dot{x}(t) = (A_n - B_f K_n) x(t) \tag{24}$$

with the initial condition $x(t_f) = \varphi(M_n, t_f) x_0$. Note that nothing can be said about the stability of the system (24).

Moreover, the accommodated control is admissible if :

$$x_0^T \varphi^T(M_n, t_f)(P_f - P_n)\varphi(M_n, t_f) x_0 \leq [\rho(x_0) - 1] x_0^T P_n x_0 \tag{25}$$

holds true. Since the left hand side of (25) is function of t_f, the system might be tolerant or not to a given fault, depending on its occurrence time. Basically, after the actuator fault, one has :

$$x(t_{ftc}) = e^{(B_n - B_f) K_n (t_{ftc} - t_f)} \bar{x}(t_{ftc}) \tag{26}$$

where $\bar{x}(t_{ftc})$ is the value the system state would have at time t_{ftc} in case no fault would have been present.

In the theoretical LQ setting, the control problem is defined on R^n and, therefore, any state $x(t_{ftc})$ is admissible, since the accommodated control recovers system stability when applied. However, this might rise difficulties in practical situations, when the control problem is defined with respect to the system operating region, possibly described by some given bounded set X delimited by matrix inequalities as in (14), since there exists the risk that $x(t_{ftc}) \notin X$, if $B_n - B_f$ and $t_{ftc} - t_f$ are not small enough, as stated by (26). Forth, the deviation during the transient trajectory might be such that, although stabilizing the system after t_{ftc} and transferring its state so that $x(t_{ftc}) \in X$, i.e. the admissibility condition (14) holds, the accommodated control leads to a non-admissible value of the cost $J_{M_n, M_f}(x_0, u_n, x_{t_f}, u_f)$, i.e. condition (13) does not hold.

Therefore, the FTC delay, $t_{ftc} - t_{fdi}$, which adds to the FDI delay, $t_{fdi} - t_f$, might imply the non-admissibility of the accommodated control and/or the system loss, but the difference $B_n - B_f$ is obviously not under control, and, besides, minimizing the quantity $t_{fdi} - t_f$ is the objective of FDI research.

D. Progressive Accommodation Scheme

The iterative scheme used in this paper aims at minimizing the incidence of the FTC computation time. It applies the intermediate control laws :

$$u_i(t) = -K_i x(t) \quad i = 1, 2, \ldots \tag{27}$$

where the control matrices are those from (16), thus taking advantage of the properties provided by the Newton-Raphson algorithm at every iteration step.

One expects this scheme to improve the faulty system behavior, without waiting for the complete convergence to the solution of the post-fault ARE in (10).

In this case, the post-fault control sequence is composed of the following steps :

1) Phase 1. First iteration :

$$\dot{x}(t) = (A_n - B_f K_n)x(t) \quad t \in [t_f, t_{fdi} + \Delta[\tag{28}$$

2) Phase 2. Progressive accommodation ($i = 1, \ldots, N-1$) :

$$\dot{x}(t) = (A_n - B_f K_i)x(t) \quad t \in [t_{fdi} + i\Delta, t_{fdi} + (i+1)\Delta[\tag{29}$$

3) Phase 3. The Newton-Raphson algorithm has converged :

$$\dot{x}(t) = (A_n - B_f K_N)x(t) \quad t \in [t_{fdi} + N\Delta, \infty) \tag{30}$$

E. Comparison between Classical and Progressive Control Schemes

In the progressive accommodation scheme, each iteration consists of solving a Lyapunov equation. Convergence speed and initialization are addressed in [27], [32], [33], where it is shown that, in addition to being monotonically convergent, the iterates are also quadratically convergent. Practically, if the number of the state variables is relatively small, convergence is rapid, with approximately 10 iterations needed in most cases. The following result is proven in [34].

Theorem 2 (Stability): Let $u_n(t) = -K_n x(t)$ be the nominal solution of the LQ problem. If $(A_n - B_f K_n)$ is stable, the progressive accommodation scheme insures the system stability on $[t_f, \infty)$ for the initialization $K_0 = K_n$. If $(A_n - B_f K_n)$ is unstable but stabilizable, there is an appropriate choice of K_0 that will insure stability on $[t_{fdi} + \Delta, \infty)$.

In the sequel, the classical and progressive accommodation schemes are applied to design a fault tolerant control sequence that addresses the loss of efficiency in the elevator actuator of a B747.

IV. B747 Longitudinal Short-Period Mode Control

In this section, a second order longitudinal control model of the Boeing 747 aircraft is considered, with the aim of controlling the short-period oscillations.

A. Flight Condition

The aircraft flight condition is assumed to be a straight and constant level flight at fixed altitude of 40000 ft and Mach number of 0.8. Considering the CG (center of gravity) position to be $(0.25, 0, 0)$ of m.a.c. (mean aerodynamic chord) and the aircraft mass to be 19792 $slug$, the corresponding steady-state speed is 774 fps. The control δ_E (rad) is the elevator deflection.

B. Nominal System

1) Open-Loop Model: Let F_B be the aircraft body-axis reference frame.

The aircraft longitudinal dynamics is described by the state vector :

$$x = \begin{bmatrix} w & q \end{bmatrix}^T \tag{31}$$

where w (ft/sec) represents the inertial velocity in the F_B z-direction and q (rad/sec) represents the pitch rate.

The linearized model of the system is :

$$\dot{x}(t) = A_n x(t) + B_n u(t) \tag{32}$$

where the system and control matrices are given in [35] :

$$A_n = \begin{bmatrix} -0.3127 & 774.0000 \\ -0.0010 & -0.4284 \end{bmatrix} \quad B_n = \begin{bmatrix} -17.9416 \\ -1.1577 \end{bmatrix} \tag{33}$$

In this flight configuration, matrix A_n has eigenvalues :

$$Sp(A_n) = \{-0.3706 \pm 0.8779\, i\} \tag{34}$$

The name "short-period" obviously follows from the aircraft natural mode having a period of 7.08 sec. It can also be observed that it is heavily damped from the magnitude of the damping ratio which is 0.389.

A 20 seconds system simulation is shown on Fig. 1, where the excitation was achieved by deflecting the control column pilot input by 5 degrees (approximately 0.0872 rad), thus applying a short doublet pulse during 3 seconds to the elevator deflection control input.

Short-period motion involves rapid changes in the angle of attack α (rad) and pitch angle θ (rad) at roughly constant speed, with the specification that, in the small-disturbance case analysis, the evolution of α and w follow approximately the same shape. One can see the natural aircraft mode evolution on Fig. 1.

2) Closed-Loop System: Let the quadratic cost functional be :

$$J(x_0, u(t)) = \int_0^\infty \left[x^T(t)Qx(t) + u^T(t)Ru(t)\right] dt \tag{35}$$

where the initial condition is $x_0 = \begin{bmatrix} 1.0000 & -0.0001 \end{bmatrix}^T$.

The trade-off between control amplitudes and settling time in the LQ problem is strongly influenced by the choice of the weighting matrices Q and R. In this application, we choose the weighting matrices as $R = 1$ and $Q = 10^{-2} I_2$.

Then, the nominal ARE :

$$A_n^T P_n + P_n A_n + Q - P_n B_n R^{-1} B_n^T P_n = 0 \tag{36}$$

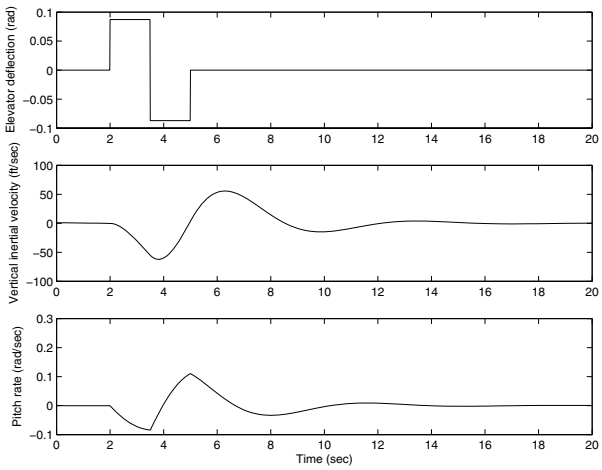

Fig. 1. B747 Short-Period Mode: Open-Loop Behavior

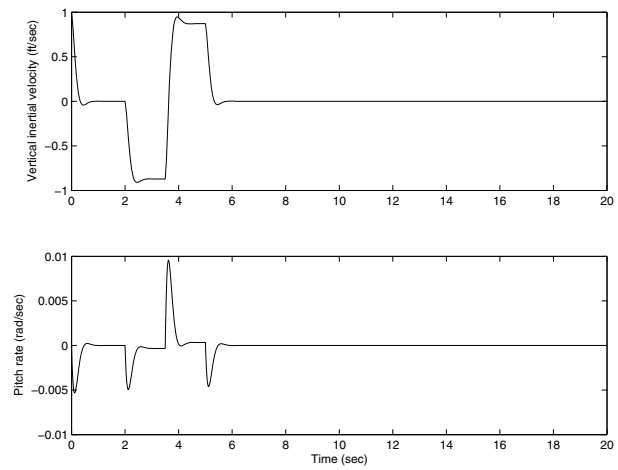

Fig. 2. B747 Short-Period Mode: Closed-Loop Behavior

with the system and control matrices as in (33), provides the solution :

$$P_n = \begin{bmatrix} 0.0012 & 0.0630 \\ 0.0630 & 7.2792 \end{bmatrix} \tag{37}$$

which yields the value of 0.0012 for the criterion in (35), and gives the linear quadratic state-feedback matrix :

$$K_n = \begin{bmatrix} -0.0954 & -9.5582 \end{bmatrix} \tag{38}$$

In the nominal case, the closed-loop system will be :

$$M_n = A_n - B_n K_n = \begin{bmatrix} -2.0236 & 602.5109 \\ -0.1114 & -11.4939 \end{bmatrix} \tag{39}$$

which is stable. Therefore, the behavior exhibited by the closed-loop system, when using (38) as nominal controller, can be seen on Fig. 2.

C. Faulty System

1) Considered Faults: The considered fault is the loss of efficiency of the actuator that controls the elevator deflection. Such faults are classically modeled by a multiplicative factor $\tau \in (0,1]$, so that the linear state-space representation of the faulty system is described by the matrices $A_f = A_n$ and $B_f = (1-\tau)B_n$. Obviously, $\tau = 0$ represents the nominal operation case, and $\tau = 1$ stands for the complete failure case, which is the situation where the actuator does not act on the system anymore.

TABLE I

ACCOMMODATED CONTROL MATRIX AFTER CONVERGENCE

τ	0.1	0.3	0.5	0.7	0.9
$k_{f_1}^{acc}$	-0.0950	-0.0942	-0.0928	-0.0900	-0.0798
$k_{f_2}^{acc}$	-10.1080	-11.5240	-13.6645	-17.4872	-27.9041
N	5	6	7	8	9

2) Progressive Fault Accommodation: After the fault, one may think of improving the system closed-loop behavior, based on the solutions provided at every iteration step by the Newton-Raphson algorithm. Even in the faulty case, using the nominal control matrix of the nominal ARE, the closed-loop system remains stable, so that K_n may be chosen as the stabilizing initial gain of the algorithm.

Convergence is reached when there are no further variations in the elements of the feedback control matrix. In Table I, for different choices of τ, one can see the number of steps until convergence, N, and the accommodated matrix $K_f^{acc} = \begin{bmatrix} k_{f_1}^{acc} & k_{f_2}^{acc} \end{bmatrix}$. Moreover, one may observe that N becomes larger as τ increases.

3) Particular Case: The previous restriction to actuator faults was made only for simplicity, but the progressive accommodation (PA) scheme obviously works for more general faults. In order to illustrate this, let us consider faults described by two coefficients, i.e. not only the control matrix $B_f = (1-\tau)B_n$, but also the system matrix $A_f = (1-\gamma)A_n$, where $\gamma \in (0, 0.1]$. The interpretation of parameter γ may be that A_n results from the linearization of a non-linear behavior around an operating point that might well be changed by the fault occurrence, or that there is a systematic error in the estimation procedure ran by the diagnostic algorithm. Consider a 90% efficiency degradation on the actuator and assume a 10% identification error of the system matrix.

The nominal control law still stabilizes the closed-loop system :

$$M_f = A_f - B_f K_n = \begin{bmatrix} -0.4525 & 679.4511 \\ -0.0119 & -1.4921 \end{bmatrix} \tag{40}$$

Hence, taking (38) as the initial gain of the Newton-Raphson algorithm, it converges to the solution :

$$P_f^{acc} = \begin{bmatrix} 0.0040 & 0.6387 \\ 0.6387 & 221.7001 \end{bmatrix} \tag{41}$$

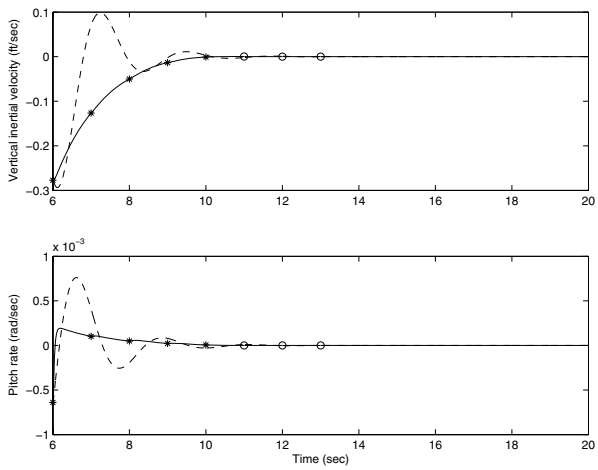

Fig. 3. B747 Short-Period Mode: Progressive (Continuous) vs. Classical (Dotted) Accommodated Closed-Loop Behavior

which gives the accommodated state-feedback matrix :

$$K_f^{acc} = \begin{bmatrix} -0.0812 & -26.8122 \end{bmatrix} \tag{42}$$

Table II shows the complete accommodation of the control matrix $K_f^{acc} = \begin{bmatrix} k_{f_1}^{acc} & k_{f_2}^{acc} \end{bmatrix}$ after $N = 8$ iterations.

Then, the accommodated closed-loop system will be :

$$M^{acc} = A_f - B_f K_f^{acc} = \begin{bmatrix} -0.4270 & 648.4947 \\ -0.0103 & -3.4896 \end{bmatrix} \tag{43}$$

which is stable.

Fig. 3 displays the simulations of the closed-loop behavior during the post-fault period, when applying the progressive and the classical control schemes, considering the sample $\Delta = 1$ *sec*. The fault occurred at $t_f = 4$ *sec*, the estimation of the faulty model was ready at $t_{fdi} = 5$ *sec*, therefore there is one sample of uncertain evolution, until $t_{fdi+\Delta} = 6$ *sec*. Only the transient, after that instant, is highlighted on the figure.

D. Admissible Closed-Loop Models

Admissibility has been defined by the two conditions (13) and (14), that respectively address the cost and the behavior of the accommodated system. As far as aircrafts are concerned, admissibility is linked with the handling qualities, that refer to those characteristics that govern the ease and precision with which a pilot is able to perform his tasks.

For each system pole, the handling qualities requirements are expressed in terms of the damping and frequency of the system mode, since these two parameters govern the system dynamic stability.

Let (ω, ζ) be the natural frequency and the damping ratio associated with the accommodated closed loop :

$$A_f - B_f K_f \triangleq \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{44}$$

To this end, it can be checked that the reference model :

$$A_n - B_n K_n = \begin{bmatrix} -0.4573 & 731.9353 \\ -0.0103 & -3.1427 \end{bmatrix} \tag{45}$$

TABLE II

CONTROL MATRIX ACCOMMODATION TILL CONVERGENCE

i	1	2	3	4
$k_{f_1}^{acc}$	-0.1778	-0.1024	-0.0783	-0.0784
$k_{f_2}^{acc}$	-223.144	-117.248	-60.4003	-35.2454
i	5	6	7	8
$k_{f_1}^{acc}$	-0.0808	-0.0812	-0.0812	-0.0812
$k_{f_2}^{acc}$	-27.6509	-26.8122	-26.8122	-26.8122

with eigenvalues :

$$\Lambda_n = \{-1.8 \pm 2.4\, i\} \tag{46}$$

provides the "best" performance recommended by the classical "thumb-print" criterion [36], as far as longitudinal short-period oscillation is concerned, i.e. the damping coefficient of about 0.6 and the frequency of approximately 3 rad/sec.

With respect to the "best" closed-loop model in (45), the sets :

$$\Phi(a,b,c,d) = \frac{(\zeta - 0.6)^2}{0.04} + \frac{(\omega - 3)^2}{0.64} - 1 \leq 0 \tag{47}$$

$$\Psi(a,b,c,d) = \frac{(\zeta - 0.35)^2}{0.49} + \frac{(\omega - 3.3)^2}{2.56} - 1 \leq 0 \tag{48}$$

define all closed loops whose dynamic performances are respectively "satisfactory" - conditions (47) and (48) both hold, "acceptable" - only condition (48) holds, or "unacceptable" - neither (47) nor (48) holds.

The performances of the nominal, post-fault, and accommodated closed-loop models, respectively in (39), (40), and (43), are displayed in Table III, which also shows those which would be obtained for another different weighting matrix Q. This table shows that, even when linear quadratic recoverability solutions exist for serious faults, they may be non-admissible.

V. CONCLUSION

Fault Tolerance (FT) represents the ability of a controlled system to still achieve the control objectives, despite the occurrence of faults, maybe at the cost of limited performance degradation. In this paper, actuator faults occurring at one single time instant have been considered, while the control objective was defined by the classical Linear Quadratic (LQ) problem on an infinite time horizon. Both fault accommodation and system reconfiguration strategies need the controller to be re-designed, which, in the LQ framework, means solving again the Algebraic Riccati Equation (ARE) associated with the post-fault system model, and checking whether the specified admissibility conditions are satisfied.

TABLE III

NOMINAL/POST-FAULT/ACCOMMODATED PERFORMANCES

Q	M_n	M_f	M^{acc}
10^{-2}	$(0.711, 9.51)$	$(0.328, 2.96)$	$(0.685, 2.86)$
10^{-4}	$(0.683, 3.01)$	$(0.414, 1.21)$	$(0.516, 1.05)$

Due to the FDI and computation delays, the classical fault accommodation scheme may involve system instability and/or control inadmissibility. These drawbacks may be limited by applying the intermediate control solutions provided by the Newton-Raphson algorithm for solving the AREs, instead of waiting for the definitive one. In this case, although the total convergence time and the computation complexity are not changed, the system behavior is improved.

The approach has been illustrated by the longitudinal short-period mode control of a Boeing 747 aircraft, under faults that decrease the efficiency of the actuator that controls the elevator deflection. The progressive fault accommodation scheme has shown to be an efficient means of dealing with the transient period, while addressing the handling qualities requirements.

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